Learning the Rhythm: Why Rotary Position Encoding Degrades Beyond the Training Horizon, and How Learnable Frequencies Help

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Abstract

Finance needs forecasts at several horizons, yet a model is usually trained for one. A Transformer that represents each future step by a placeholder token, told apart only by rotary position encoding (RoPE), can be trained once and queried at any horizon — but with standard RoPE, trained at 96 steps and asked for 720, its error reaches 6.4 times that of a model trained at 720. A learnable scale on each head's RoPE frequencies (4 parameters per layer) cuts this error by 38–66% on seasonal benchmarks and 16–38% on two electricity markets. We trace the effect to phase drift: RoPE's fastest rotation is out of step with the daily cycle, so far-ahead positions carry the code of a different hour, and attention moves to exactly the hour that RoPE arithmetic predicts. Fixing the day-aligned scale reproduces the gain; resetting the learned scales removes it.

1 Problem, motivation and related work

Risk managers report Value-at-Risk at 1 and 10 days, HAR volatility models combine daily, weekly and monthly components [1], and energy traders bid day-ahead and hedge the next week [8]. Deep long-horizon forecasters, however, are trained *per horizon* [9, 5]; a model trained once and queried at any horizon would remove this cost. Our proposal gave each attention head a learnable multiplier on its RoPE frequencies [7], expecting heads to specialise to time scales like HAR's components. At the training horizon this was not supported: the scales stayed within 10% of 1, heads learned similar values and accuracy barely changed. The same scales, however, mattered greatly *beyond* the training horizon. **Contributions:** (i) we document a silent failure of RoPE forecasters beyond the training horizon; (ii) four learnable scales per layer largely fix it; (iii) we derive the mechanism — phase drift relative to the daily cycle — and verify its prediction in the attention weights. **Related work.** Transformer forecasters [10, 5] are evaluated at fixed horizons, and linear models are strong baselines [9]. RoPE length extrapolation is studied for language models through frequency interpolation and rescaling [6]; text has no period, whereas time series do, so the relevant question becomes whether RoPE's rotation is commensurate with the calendar.

2 Model and method

Backbone (Fig. 1). A univariate history of L=336 steps is cut into 21 patches of 16 steps, projected to d=64, and followed by $\lceil H/16 \rceil$ placeholder tokens that all equal one learnable vector m. A 2-layer, 4-head pre-norm encoder processes the sequence and a shared linear layer reads each placeholder into 16 forecast values. With the mask of Fig. 1, history representations do not depend on H and step h's forecast is identical for any $H \geq h$ (unit-tested), so a model trained with 6 placeholders (H=96) can be queried with 45 (H=720). We chose patches because point tokens gave much noisier extrapolation, and placeholders (rather than a flat output layer) because only they make the horizon a model input; RoPE is the only position signal.

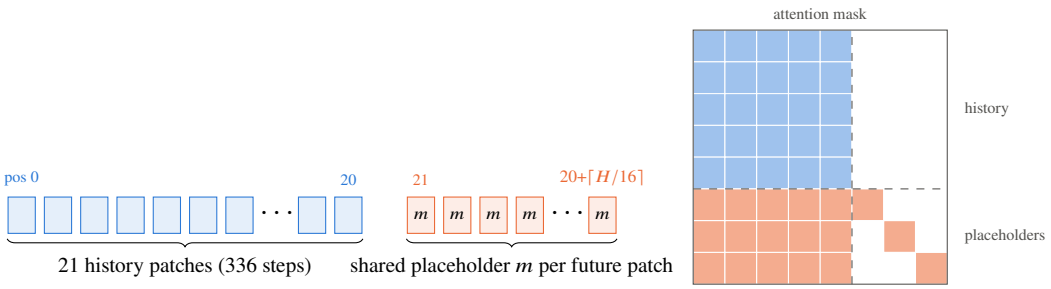


Figure 1: Forecaster. Left: history patches and one shared placeholder per future patch; only RoPE positions distinguish placeholders. Right: mask (coloured = allowed): history sees history; a placeholder sees history and itself, so the forecast for step h does not depend on H.

RoPE. Pair j of a head's query/key coordinates is rotated by $t\omega_j$ at position t, $\omega_j = b^{-j/8}$ ($b \in \{10^3, 10^4\}$, 8 pairs). Since $R(\alpha)^\top R(\beta) = R(\beta-\alpha)$, the attention logit between positions t and $u = t+\Delta$ is

$$s(\Delta) = \frac{1}{\sqrt{d_h}} \sum_j \left[a_j \cos(\omega_j \Delta) + c_j \sin(\omega_j \Delta)\right], \tag{1}$$

with content-dependent a_j, c_j: attention sees only the lag, and each pair adds a sinusoid of it. The fastest pair turns at ω_0=1 rad per token for every base.

Learnable scales (our method). Each layer ℓ and head h uses $\hat{\omega}_{\ell,h,j} = \omega_j e^{a_{\ell,h}}$ with $a_{\ell,h}$=0 at initialisation (exactly standard RoPE), trained jointly without weight decay: 4 parameters per layer, no extra compute. Ablations use the same backbone with *grouped* scales (fast and slow pairs separately), *shared* scales (one per layer) or *fixed* hand-set scales.

Why alignment matters. Let one day be T tokens. Lags Δ and $\Delta + nT$ are the same time of day, and the fastest pair encodes them with angles differing by $n\hat{\omega}_0 T$.

Proposition 1. *If $\hat{\omega}_0 T = 2\pi k$ ($k \in \mathbb{Z}$), the fastest pair gives Δ and Δ+nT identical codes for all n; otherwise each day rotates the code by the drift $\delta = \mathrm{wrap}(\hat{\omega}_0 T)$.*

Training only shows placeholders lags up to 26 tokens; beyond them, with drift, a same-hour lag carries the code of a *different* hour seen in training, and a model that learned "look at the same time yesterday" should attend to that wrong hour. On ETTm1 (15-min data) one day is T=6 tokens: standard RoPE drifts by $6-2\pi \approx -16°$ per day and the aligned scale is $2\pi/6 \approx 1.047$. **Multi-horizon training**, a second remedy, draws each batch's horizon from {96, 192, 336, 720} so distant placeholders are seen in training; unlike the scales, it needs 720-step targets.

3 Data and experimental protocol

Data (URLs in Appendix B). Four benchmarks — Electricity (321 series, hourly), ETTh1/ETTm1 (7 series, hourly/15-min), Weather (21, 10-min) — and three financial markets chosen before seeing results: hourly day-ahead prices of Nord Pool and PJM (EPF-NP/PJM) [3], which have daily cycles, and daily exchange rates of 8 currencies, which do not [4]. Each variable is forecast as its own series; splits are chronological (ETT 12/4/4 months, others 70/10/20%); z-scores use training rows only; exchange rates get per-window normalisation (RevIN), without which every model fails.

Protocol. All models read L=336, train at H=96 and are tested at every step to 720 on the same windows; variants share backbone, optimiser, step budget, batch order and initial weights, differing only in RoPE frequencies. RoPE base and checkpoint are chosen on validation at H=96; the test set is never used for training or tuning. **Baselines:** fixed RoPE (isolates the mechanism), a linear map $\mathbb{R}^{336} \to \mathbb{R}^H$ (strong on these data [9]), and models trained directly at 720. **Significance:** with 3 seeds $\times$ 2 bases we count improving runs, and within each run apply a Diebold–Mariano test [2] to the per-origin loss difference on steps 97–720 (Newey–West, lag 719, for overlapping windows). Metric: MSE on the normalised scale.

Table 1: Train and test MSE at the training horizon H=96 (mean±std over seeds). PatchTST/42 reports test 0.130, 0.375, 0.290, 0.152 on Electricity, ETTh1, ETTm1, Weather ([5], Table 3).

	Linear		Transformer, fixed RoPE		Transformer, learnable	
Dataset	train	test	train	test	train	test
Electricity	0.139±0.000	0.140±0.000	0.133±0.001	0.139±0.001	0.133±0.001	0.139±0.001
ETTh1	0.336±0.001	0.377±0.002	0.338±0.013	0.417±0.017	0.340±0.010	0.414±0.011
ETTm1	0.272±0.003	0.304±0.004	0.223±0.010	0.311±0.009	0.219±0.008	0.317±0.011
Weather	0.454±0.001	0.174±0.000	0.388±0.006	0.153±0.000	0.388±0.004	0.153±0.001
EPF-NP	0.222±0.000	0.472±0.000	0.149±0.014	0.482±0.032	0.139±0.006	0.452±0.013
EPF-PJM	0.637±0.001	0.163±0.000	0.235±0.008	0.174±0.015	0.378±0.143	0.167±0.005
Exchange	0.122±0.000	0.089±0.000	0.120±0.022	0.103±0.010	0.114±0.022	0.099±0.005

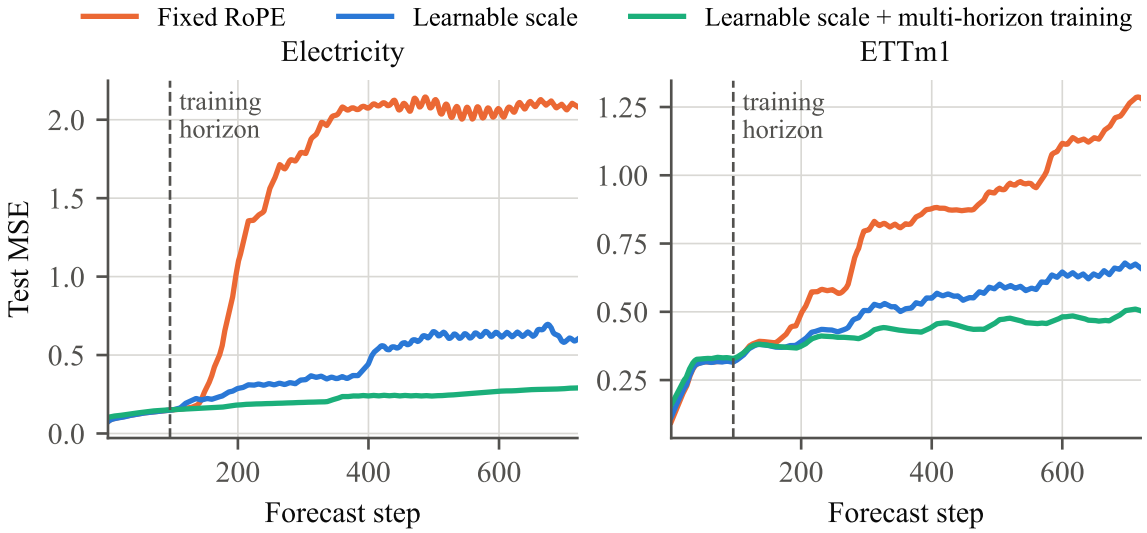


Figure 2: Test MSE per forecast step, trained at H=96 (base 10^4, 3 seeds).

4 Results

At the training horizon (Table 1) the backbone is close to PatchTST; it beats the linear model on Weather and EPF-NP, ties on Electricity and loses on ETTh1, ETTm1 and exchange rates, as reported by [9]. Fixed and learnable RoPE differ by about one seed standard deviation or less (Weather's test period is calmer than its training period).

Beyond the training horizon (Fig. 2, Table 2) fixed RoPE degrades within a day or two and reaches 1.5–6.4 times the error of a Transformer trained at 720 on Electricity, ETTh1 and ETTm1. Learnable scales reduce the 720-step error by 66% (Electricity), 51% (ETTh1) and 38% (ETTm1), significantly in all 6 runs each; on ETTh1 the extrapolating model even beats both models trained at 720, which overfit. On Weather and exchange rates the effect is small and of mixed sign.

Mechanism. Four results support the explanation of Sec. 2. *(1) Head diversity is not needed:* one scale shared by all heads works as well as one per head, and hand-set spreads across heads help little or hurt (Table 3) — contrary to our original proposal. *(2) The effect sits in the fast pairs:* grouped scaling equals per-head scaling; its slow group drifts (0.95–1.13) without effect, as slow pairs barely change from day to day. *(3) The learned value is the aligning value, and it is necessary:* on ETTm1 the scales move from 1 towards 1.047 (mean 1.036, Fig. 4a); fixing 1.047 without learning gives the same gain, 0.955 or 1.10 do not (Fig. 3; 1.047 beats 0.955 in all 6 runs, DM $p < 10^{-10}$), and resetting a trained model's scales to 1 at test time brings the failure back (Fig. 4b). On Weather (9 tokens per day) fixing $2\pi/9$ likewise lowers the 720-step error (0.359/0.366 vs 0.392/0.377 at bases $10^3/10^4$). *(4) Attention moves where RoPE arithmetic predicts* (Fig. 5): placeholders look up the same time of day;

Table 2: Test MSE over steps 1–720. Trained at 96 and queried at 720 (fixed vs learnable); DM: runs of 6 significantly better / worse with learnable scales (p<0.05); one multi-horizon model; models trained at 720.

	Trained at 96		DM	Multi-horizon	Trained at 720	
Dataset	fixed	learnable	+/−	learnable	Transformer	Linear
Electricity	1.347±0.087	0.464±0.107	6/0	0.216±0.003	0.210±0.002	0.204±0.000
ETTh1	0.972±0.071	0.473±0.013	6/0	0.541±0.014	0.632±0.036	0.493±0.004
ETTm1	0.764±0.068	0.474±0.019	6/0	0.421±0.011	0.435±0.013	0.430±0.003
Weather	0.392±0.009	0.396±0.006	2/1	0.326±0.009	0.336±0.004	0.325±0.001
EPF-NP	1.072±0.061	0.896±0.039	5/0	1.123±0.008	1.196±0.156	0.863±0.003
EPF-PJM	0.443±0.036	0.273±0.034	6/0	0.216±0.002	0.213±0.003	0.221±0.000
Exchange	1.078±0.100	1.024±0.036	3/3	1.074±0.087	1.097±0.214	0.890±0.003

Table 3: Mechanism ablations (test MSE, trained at H=96).

Frequency scales	Test MSE
ETTh1, steps 1–336, 5 seeds	
Fixed RoPE (scale 1)	0.676±0.022
Learnable, per head	0.460±0.006
Learnable, grouped	0.460±0.006
Learnable, shared by heads	0.458±0.011
Fixed, mild spread (0.9–1.1)	0.547±0.038
Fixed, wide spread (0.5–2.0)	0.716±0.064
ETTm1, steps 1–720, 3 seeds × 2 bases	
Fixed RoPE (scale 1)	0.767±0.051
Learnable, per head	0.488±0.033
Fixed 1.047 (= 1 day)	0.459±0.013
Fixed 0.955 (misaligned)	0.923±0.065
Fixed 1.10 (misaligned)	0.809±0.044

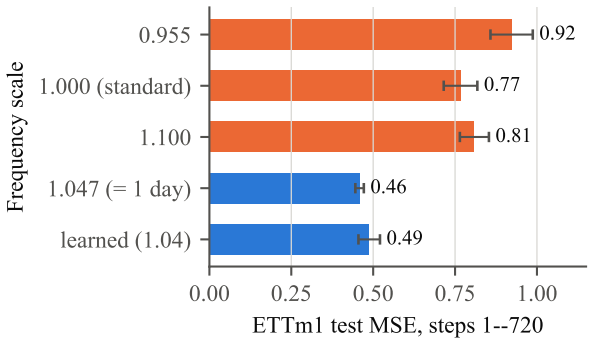


Figure 3: ETTm1: scale $2\pi/6$=1.047 aligns the fastest rotation with the day; fixed at 1.047 it matches the learned model, 5% off it fails.

with fixed RoPE this share falls from 0.66 to 0.03 over 7.5 days while attention shifts to +4 h and then +8 h — the offsets whose codes best match training lags, computed from the angles alone — whereas with 1.047 it stays (0.49). Gains also grow with seasonality: in Weather only the three solar-radiation channels gain, and exchange rates, with no cycle, show none. On hourly data (1.5 tokens per day) gains are as large but the learned scales (1.0–1.07) are not the day-aligned value; which cycle is matched there remains open.

Multi-horizon training closes most of the remaining gap: one model is within 3% of, or better than, the Transformer trained at 720 on all seven datasets (Table 2), at a small cost at H=96 (Electricity 0.135 vs 0.128). It is the stronger remedy when 720-step targets exist; learnable scales help when they do not or when long-horizon training overfits (ETTh1, EPF-NP).

5 Application to finance

Electricity prices are traded with a strong daily profile; learnable scales cut 720-step error on both markets (EPF-PJM 0.443→0.273, significant in 6/6 runs; EPF-NP 1.072→0.896, 5/6). Exchange rates behave like a random walk: the scales have no consistent effect (3 runs better, 3 worse) and the linear model is best. Practical rules: (i) for seasonal series — intraday volume and volatility with their daily U-shape, power and gas prices — use learnable or calendar-aligned RoPE ($\hat{\omega}_0 = 2\pi/T$ for the tokenisation used) before forecasting beyond the training horizon; (ii) to serve several risk horizons (1- and 10-day VaR, HAR-style volatility) with one model, train on mixed horizons; (iii) for near-random-

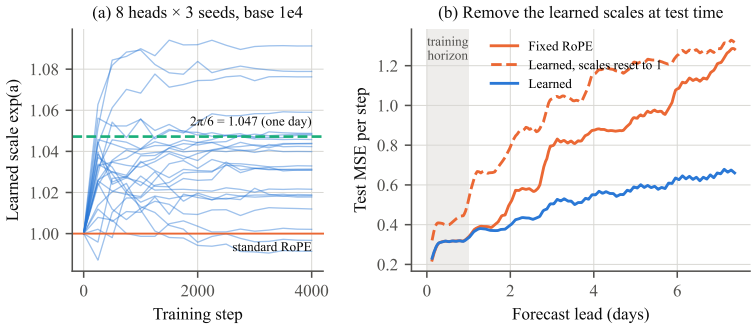


Figure 4: ETTm1, base 10^4. (a) Learned per-head scales during training (8 heads × 3 seeds); dashed: day-aligned $2\pi/6$. (b) Per-step test MSE (6-hour average) of fixed RoPE, the learnable model, and the same weights with scales reset to 1 at test time.

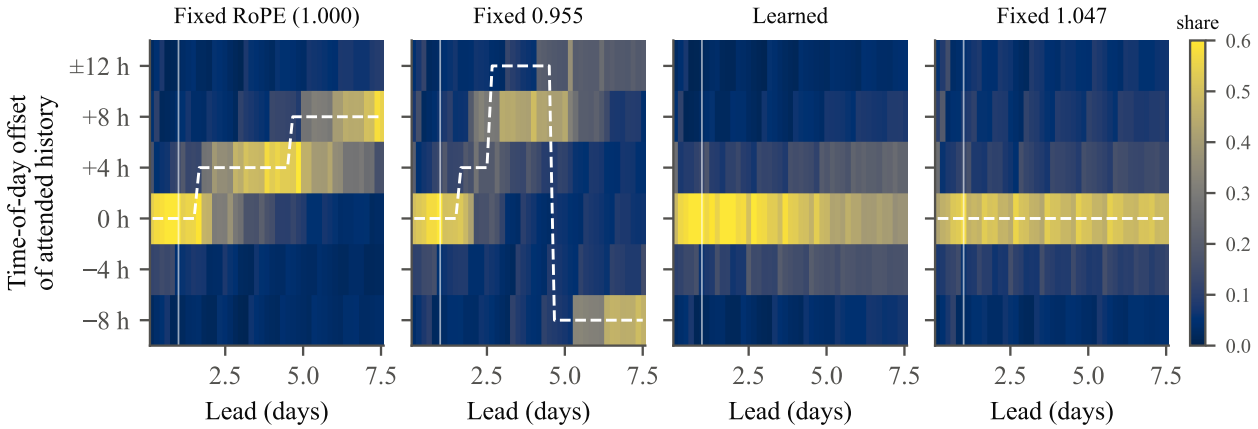


Figure 5: Attention of placeholders by time-of-day offset of the attended history (ETTm1, queried at 720; last layer, 3 seeds). Dashed: offset predicted from RoPE angles alone.

walk prices, position encodings did not help.

6 Limitations and conclusion

Heads did not specialise to time scales and grouping gave no gain; the model is small (2 layers, d=64), seeds are few (3–5, partly offset by DM tests), and mechanism analyses cover ETTh1, ETTm1 and Weather. RoPE degrades beyond the training horizon because its rotation is out of step with the daily cycle; four learnable scales per layer realign it, and mixed-horizon training narrows the rest. A position encoding is a clock, and it must keep the market's time.

References

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Appendix A. Step-by-step replication

All runs used Python 3.13 and PyTorch 2.6 (CUDA 12.4) on one NVIDIA H100 (memory capped at 8%); a run takes 20 s–3 min. Code: `https://github.com/ycsama0703/ft5011-ropeformer`, branch `final-submission` (setup and all commands in `README.md`). **Trained models and all result files** (257 checkpoints, 71 MB) are attached to the release `https://github.com/ycsama0703/ft5011-ropeformer/releases/tag/final-report-2026-11-10`; unzipped at the repository root they reproduce every table and figure without retraining. Retraining ETTh1 from scratch on a different GPU (RTX 3090, 12 runs) gave a 720-step MSE of 0.982 (fixed) and 0.463 (learnable) against 0.987 and 0.472 in the report; single runs differ by up to 3% because GPU kernels are not deterministic.

Table 4: Hyper-parameters (identical for every RoPE variant) and datasets.

Lookback / patch	336 steps / 16 (21 tokens)	Optimiser	AdamW, lr 10^{-3}, wd 0.01
Width / heads / layers	64 / 4 / 2	RoPE scales	no weight decay
Feed-forward / dropout	128 / 0.1	Schedule	300 warm-up steps, cosine
RoPE base	10^3 or 10^4 (validation)	Clip / batch / precision	1.0 / 256 / bf16
Training horizon	96 (6 placeholders)	Step budget	2000 ETTh1, EPF, exchange;
Evaluation horizon	720 (45 placeholders)		4000 ETTm1; 6000 Weather; 8000 Electricity

Dataset	Source	Freq.	Length	Series	Split
Electricity	Time-Series-Library	hourly	26,304	321	70/10/20
ETTh1 / ETTm1	Time-Series-Library	hourly / 15 min	17,420 / 69,680	7	12/4/4 months
Weather	Time-Series-Library	10 min	52,696	21	70/10/20
EPF-NP / EPF-PJM	Lago et al. (2021), price only	hourly	52,416	1	70/10/20
Exchange rates	Time-Series-Library	daily	7,588	8	70/10/20, RevIN

1. `pip install -e .`; `python -m unittest discover -s tests`.
2. Data: `python tools/download_data.py ETTh1 ETTm1 weather electricity exchange_rate`; financial data: `python experiments/build_fin.py raw data` after downloading EPF (Appendix B).
3. Train at 96 and extrapolate, with direct-training references, for dataset `D` and step budget `S`: `python tools/extrap_study.py --dataset D --steps S --patch-len 16 --amp bf16 --seeds 0 1 2 --bases 1000 10000 --modes standard head grouped --eval-horizons 192 336 720 --direct-horizons 336 720 --direct-modes head --out results/r2_D.json` (add `--instance-norm` for exchange rates).
4. Multi-horizon: `python tools/multi_horizon.py --dataset D --steps S --train-horizons 96 192 336 720 --test-horizon 720 --out results/mh_D.json`.
5. Ablations (Table 3): `extrap_study.py` with `--modes shared_group fixed_head:0.9,0.97,1.03,1.1 fixed_head:0.5,0.8,1.25,2.0` (ETTh1, 5 seeds) and `fixed_head:1.0472,1.0472,1.0472,1.0472` etc. (ETTm1, Weather with 0.6981).
6. Train/val/test MSE, linear reference, DM tests, per-channel analysis and seasonal strength: `experiments/` (`train_eval.py`, `linear96.py`, `dm_test.py`, `per_channel.py`, `periodicity.py`; commands in `experiments/README.md`).
7. Tables and figures: `python report/make_assets.py results`; attention and scale figures from the released checkpoints (about 10 s on CPU): `python report/mechanism_figures.py --results results --data-dir data`; phase arithmetic: `python report/phase_figures.py`.

Appendix B. Data sources

- ETTh1, ETTm1, Weather, Electricity, exchange rates: Time-Series-Library collection (CC BY 4.0), `https://huggingface.co/datasets/thuml/Time-Series-Library`.
- EPF day-ahead prices (Nord Pool, PJM; hourly 2013–2018): `https://zenodo.org/records/4624805` (files `NP.csv`, `PJM.csv`); we keep the price column only.